

Drone Ballet - Questions for Dr. Bhaskaran

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Questions Pertinent to Research Project

- What would be a reasonable order of abstraction for the overall experiment design?
- What is the best way to look at convergence given a mesh size Δx and time-step size Δt ? What are the ways to optimize computational efficiency (e.g. bigger time-steps, bigger mesh size)?
- What is the best software for transient heat transfer behavior without a non-Cornell license?
- Can the mesh of a body given temperature gradients be saved and passed into ANSYS?
- What are the easiest ways to speed up computation in Matlab? Making everything a matrix? Using more cores?
- What would a reasonable stopping point be for a heat transfer analysis? An independent part's plastic deformation?
- Is it worthwhile to model the downwash of a drone using fluid dynamics? Or, would it be appropriate to use empirical correlations?